

Leveraging Unlabeled Data for Lung Sound Classification Through Self-Supervised Contrastive Learning

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Abstract

The use of Deep Learning (DL) in biomedicine has attracted recent interest in the scientific community because of its ability of generating more complex and heterogeneous data processing models compared to those previously based on traditional machine learning techniques. In the healthcare domain, its application to biomedical image processing is particularly noteworthy. Nevertheless, its utilization on biomedical audio signals is not as widespread. In this context, this paper focuses on the classification of lung sounds as can be instrumental in diagnosing respiratory diseases, that are one of the leading causes of death. In particular, we address two intrinsic challenges associated with the application of DL models in the health field. The first one is the need of large databases annotated by expert medical staff, thus limiting the generation of such models due to the heavy manpower resources required. The second one is the presence of a strong class imbalance in the dataset primarily due to the predominance of non-pathological respiratory sounds. In this paper, for overcoming these issues we propose a DL system based on the contrastive self-supervised learning paradigm, in particular the

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SimCLR technique, that is able to generate useful audio data representations from unlabeled data. Results show that these self-supervised audio representations can be effectively transferred to the task of interest using a limited amount of annotated data, outperforming the conventional supervised learning model when the amount of available labeled data is scarce. In addition, the proposed system is able to reduce the bias towards the majority class, improving the sensitivity of the fully supervised model. We believe that the findings in our paper enlightens the path towards the use of unlabeled data in biomedicine alleviating the need of large annotated datasets.

Keywords: Respiratory Sound Classification, Wheeze, Crackle, Unlabeled Data, Class Imbalance, Self-supervised Learning, Contrastive Learning, Transfer Learning, Data Augmentation

1. Introduction

Deep Learning (DL) architectures and methods have been of great interest in recent years due to their ability to generate complex models and process heterogeneous data types. One of the fields where these algorithms are being proposed with great success is medicine. Prior to the surge in neural network-based models, task-specific statistical models and traditional Machine Learning (ML) algorithms were commonly used. However, the main drawback of this approach is the need to perform a feature engineering process to extract and transform the raw data into variables suitable to the task in hand.

The capability of deep networks to extract those features by themselves have turned them into the focus of models that work with more complex

data like images, audio, video, language and graphs. In the area of health-care applications, although there is a substantial amount of literature on the subject, most of it focuses on biomedical image processing, as for example, in (Roth et al., 2014; Rajasekar et al., 2023) where these techniques have been proposed for detecting the early features of cancer using 3D Computerized Tomography (CT) scanings of the chest. However, voice and body sound signals can also be utilized in the biomedical field as a large number of existing diseases produce disturbances in them. In fact, this is an emerging topic of study though it is not yet as widespread as medical image processing.

In this work, we delve into the use of audio signals for biomedical applications leveraging on the DL capabilities. In particular, we focus on the task of lung sound classification, whose objective is to automatically categorize recordings of respiratory cycles in four different classes: normal sounds and three different types of abnormal sounds (wheezes, crackles or both). Although the auscultation is the gold-standard in this task, it is dependent on the knowledge and skills of the medical staff. Therefore, it is essential to research and develop automatic and objective methods to aid in the diagnosis of respiratory diseases. In addition, it is expected that these solutions have a great social and medical impact as over the decade from 2010 to 2019, respiratory illnesses take a place in the top 10 of the leading sources of death according to the World Health Organization (WHO, 2020).

First proposed systems for lung sound classification extracted hand-crafted acoustic features that were later classified by using Empirical Ruled-Based (ERB) algorithms or traditional machine learning methods. Systems based on the first approach were based on creating classification thresholds from

some prior knowledge or characteristics of the abnormal sounds of interest (Liu et al., 2015). In the second case, it was common to use spectrum-based features (as for example, the well-known Mel-Frequency Cepstrum Coefficients, MFCC) as input to conventional ML algorithms, such as Hidden Markov Model (HMM) for normal and abnormal breath sound classification (Matsunaga et al., 2009), and Random Forest (RF) (Mendes et al., 2015) and Support Vector Machine (SVM) (Bokov et al., 2016) for wheeze detection. Other approaches extract higher-order statistics of the lung sounds that are further processed by a K-Nearest Neighbors (KNN) model (Naves et al., 2016).

As mentioned before, the main drawback of these methods is the need of the explicit design of ad hoc features suitable for the problem to solve. For that reason, researchers have begun to consider the idea of incorporating modern deep learning architectures for automatically extracting more complex and relevant features. The most commonly approach is converting the audio signal to a 2D time-frequency representation (typically, spectrogram or mel spectrogram), that can be considered as an image, and later applying convolutional operations over it through Convolutional Neural Networks (CNN). An early attempt of this procedure was proposed for wheeze detection in (Kochetov et al., 2017), achieving results that surpassed previous ML models on that task. Recent approaches continue to use CNNs, such as in (Kumar et al., 2024), where a multimodal system based of CT images and cough/lung sounds is proposed for the binary diagnosis of Chronic Obstructive Pulmonary Disease (COPD) patients. CNNs emerged inspired by the neural hierarchical processing of the brain in living organisms vision, ex-

plaining its predominant use in image processing and computer vision tasks. Compared to images, audio signals are not static since they have a temporal dimension. Taking it into account, others researchers considered using Recurrent Neural Networks (RNN) to capture temporal feature dependencies, as in (Kochetov et al., 2018) where this strategy was evaluated for multi-class abnormal respiratory sound detection. More recent approaches focus on improving those architectures or even combining them creating hybrid models (Petmezas et al., 2022).

All the previous methods rely on Supervised Learning (SL), meaning they need labeled data to build the models. In the case of DL, this requirement is especially relevant because these methods can outperform traditional ML techniques, at expenses of a larger amount of annotated data. This drawback is aggravated in specialized scopes, such as healthcare, where the labeling process must be performed by specialized personnel (in our case, pulmonologists), that then poses a challenge to the generation of large biomedical databases. In addition, in biomedicine, it is usual to have datasets with skewed class proportions due to a high variability of frequency in pathologies. Class-imbalance data is a serious problem in supervised DL systems, as the common function to be minimized in the iterative SL process (the cross-entropy loss) tends to generate poor separability margins between classes in the representation space and bias towards majority classes (that, in our case is the normal lung sound). In this context, Kang et al. (2021) shows a significant downgrade in performance when artificially increasing the imbalance of the dataset labels. In addition, Zhang et al. (2023) reviews deep learning approaches in addressing long-tailed distribution learning, showcasing a

considerable decrease in performance from head to tail classes when using cross-entropy loss compared to some proposed balancing approaches.

In the particular case of lung sounds discrimination, some solutions have been proposed in order to mitigate the need of large annotated datasets. One notable approach involves utilizing open-access pre-trained models, such as the VGG16 model (Demir et al., 2020b). In addition, the problem of skewed class distribution has been addressed in (Salor-Burdalo and Gallardo-Antolin, 2022) through the application of the mixup data augmentation technique to artificially increase the representation of minority classes. Similarly (Petmezas et al., 2022) proposes the use of the focal loss as an alternative to the conventional cross-entropy loss during the training process.

Self-Supervised Learning (SSL) (Gui et al., 2024) is a recent paradigm that has emerged as a potential solution to the downsides of SL. SSL does not rely on labeled datasets, as it autonomously generates pseudo labels to serve for supervision in predefined pretext tasks, such as predicting missing parts of images or words in a sentence. This process produces a complex and meaningful representation space that can be transferred to other applications of interest, also called downstream tasks, and further refined with minimal labeled data. Contrastive Learning (CL) (Jaiswal et al., 2021) is one of the most employed methods for SSL due to its simplicity and good performance. The main idea is to build a mapping function that preserves the similarities on the input data in the output representation space (embedding space). In other words, CL seeks to bring similar exemplars closer together (a sample and its “positive” instance) while pushing dissimilar exemplars farther apart (“negative” instances). In the absence of labels, a positive pair typically

comprises data augmentations of the same sample, while negative pairs are a subset of the remaining ones. The mapping function is learnt by minimizing a certain contrastive loss instead of the conventional cross-entropy function used in supervised learning. One of the most commonly used variants of CL is SimCLR (Chen et al., 2020), that was initially proposed for developing visual representations in computer vision tasks.

Regarding the data imbalance issue, there is an ongoing research into the capabilities of SSL for generating balanced representation spaces even in highly skewed datasets. In this context, (Assran et al., 2022) proves that SSL joint-embedding-methods, e.g. SimCLR, can enhance the quality of the learned representations on these scenarios. (Yang and Xu, 2020) show that the model benefits from adding extra unlabeled data in a semi-supervised manner in such a way that self-supervised pre-training outperforms the baseline models. Besides, (Kang et al., 2021) compares cross-entropy and contrastive loss under heavily imbalanced scenarios, revealing that the second one is more stable than the first one when increasing the imbalance of the dataset and exhibits a better generalization in cross-domain and cross-task settings. They also develop a metric to measure the balancedness of the generated feature space.

Most research work on SSL has been carried out on image processing and computer vision considering well-known large datasets like ImageNet, with much less exploration in audio-based tasks. One of the pioneer studies in this topic is (Al-Tahan and Mohsenzadeh, 2020) where it is addressed the potential of SSL CL, in particular, SimCLR, for learning auditory representations in different benchmark audio domains (speech, music and environmental

sounds).

Moreover, there are far fewer studies dedicated to the use of SSL CL for respiratory sounds in the field of biomedical applications. (Soni et al., 2022) proposes a SSL SimCLR algorithm that employs patient metadata to generate the representations instead of the usual data augmentations in CL. The method is applied to a different and easier task to ours, that is binary patient diagnosis as healthy or non-healthy. In (Song et al., 2021) and (Moummad and Farrugia, 2023), it is explored the use of supervised contrastive learning for solving the same problem as ours. However, these works differ from ours in that they combines the strengths of the SL and CL frameworks at the expenses of requiring labeled data in the CL process. Therefore, they follow a constrastive learning approach but without self-supervision. In addition, in (Moummad and Farrugia, 2023) metadata is utilized in a similar fashion to (Soni et al., 2022). In spite of the interest of the use of some kind of metadata information, it presents the drawback that is not always available or easy to employ due to data protection regulations. Finally, it is worth mentioning that these studies do not delve deeply into the behaviour of SSL CL methods in scenarios with limited and imbalanced labeled data nor in absence of available metadata in the context of multi-class classification of respiratory sounds.

The primary contribution of this paper lies in the exploration of self-supervised learning methodologies, with a particular focus on contrastive learning, for the classification of respiratory sounds. This study addresses the challenges posed by small datasets that are commonly encountered in biomedical applications: limited number of labeled instances and significant

class imbalance. This can be summarized in the following points:

- We have investigated the potential of self-supervised contrastive learning, particularly SimCLR, to generate a deep learning encoder capable of extracting audio representations from unlabeled data that are meaningful for the task of respiratory sound classification.
- We have evaluated the impact of various data augmentation techniques employed during the training of the encoder on the quality of the resulting representations for the downstream task of respiratory sound classification.
- We have compared the performance of the conventional supervised learning approach and the proposed self-supervised learning system on different labeled training data set sizes for our task of interest, showing the good performance of our approach when a small amount of annotated data is available.
- We have shown the capability of SSL to produce a more balanced representation space compared to SL in conditions of labeled data scarcity, thereby enhancing the sensitivity of the respiratory sound classification system.
- We have evaluated the performance of the SSL model against an open-access, pre-trained audio DL model trained on an extensive external dataset, showing the superiority of our approach for our target task.

This paper is organized as follows: Section 2 describes the database of lung sounds used in our experimentation. Section 3 presents the main as-

pects of the methodology followed in this work, including the description of the two approaches considered for the design of the respiratory sound classification system: supervised learning (baseline), and self-supervised contrastive learning (our proposal). Section 4 presents the experiments and results, to end with some conclusions in Section 5.

2. Database Description

The dataset considered in this study is the lung sound database presented at the ICBHI 2017 (Rocha et al., 2019) conference in a framework of a challenge devoted to the evaluation of algorithms regarding respiratory sound diagnosis. It consists of 5.5 hours of recordings of respiratory cycles from 126 patients, adding up to 920 audio files. These recordings have been resampled to a common sampling frequency (16 kHz , in this case), and manually divided into respiratory cycles annotated into four different acoustic classes: normal, crackles, wheezes, and simultaneous wheezes and crackles. Wheezes and crackles are the two most significant abnormal lung sounds. Whereas wheezes are continuous, high-pitched sounds caused by airway obstruction (Meslier et al., 1995), crackles are explosive, discontinuous sounds that occur during both the inspiratory and expiratory phases of breathing, and they last for a much shorter duration compared to the overall breathing cycle (Piiirilä and Sovijärvi, 1995).

In total, the database contains 6,898 audio excerpts corresponding to 3,642 normal respiratory cycles, 1,864 crackles, 886 wheezes and 506 cycles containing both crackles and wheezes, resulting in a highly imbalanced dataset as the normal class accounts for more than 50% of the audio samples.

The respiratory database also contains a diagnosis for each patient who participated in the recording: healthy, chronic obstructive pulmonary disease, asthma, bronchiectasis, lower respiratory tract infection, bronchiolitis, upper respiratory tract infection (URTI) and pneumonia. Some diagnoses like asthma are only registered once. It also contains extra demographic information of the patients like age and sex. In addition, every recording has extra metadata regarding the location on the chest where it was captured, the acquisition mode (sequential or simultaneous) and the recording equipment, enriching the potential information of the database.

The main goal of our study revolves around the classification of the respiratory cycles using exclusively the audio recordings as in the ICBHI competition, so that information related to patient diagnosis, demographic data and metadata is not considered. The ICBHI challenge proposed a holdout evaluation with fixed training and test sets. However, in order to obtain more reliable results and following previous works, such as (Petmezas et al., 2022), in our experiments we have adopted a cross-validation strategy as explained in Section 4.1.

3. Methodology

The aim of this paper is to propose a self-supervised approach for the task of lung sound classification and show its advantages with respect to the conventional supervised strategy regarding performance (accuracy) and ability to deal with small and highly class-imbalance datasets.

As both alternatives share some components in common, such as audio preprocessing, techniques to handle imbalanced data in the training process

and data augmentation, in this Section, firstly, we describe these modules and secondly, we focus on the specific details regarding the supervised and self-supervised approaches.

3.1. Audio preprocessing

Both, the supervised and self-supervised models are based on a 2D-CNN architecture that automatically learns the more appropriate features for the task under consideration. In order to accommodate the input to the requirements of this convolutional part, a preprocessing step of the raw audio data needs to be performed. In particular, following the state of the art in audio processing and previous studies on respiratory sounds classification using CNNs (Demir et al., 2020a; Song et al., 2021; Petmezas et al., 2022; Moummad and Farrugia, 2023), the audio was transformed to a time-frequency representation, the mel-spectrogram. Before computing it, the duration of the raw audio data was set to 8 s as in (Moummad and Farrugia, 2023), in order to obtain mel-spectrograms with the same size independently of the length of the original audio signal. In case of a longer duration, the first 8 s were trimmed, and otherwise, a padding was applied. The padding effects chosen were zero padding or cyclic padding up to 8 s long. After a preliminary experimentation, the first type of padding was selected in the self-supervised pre-training process (i.e., when no labels were used), whereas the circular one was employed in the training stage of the conventional supervised approach and in the fine-tuning of the pre-trained models (i.e., when labeled data was employed).

The parameters used to generate the mel-spectrograms are the same that have been used in (Moummad and Farrugia, 2023): a window and hop length

of respectively, 1024 and 512 samples, and a mel-scale filterbank composed of 64 triangular filters. The band of frequencies considered is between 50 Hz and 2000 Hz as previous studies have observed that most of the spectral components of wheezes and crackles are located in this range (Jakovljević and Lončar-Turukalo, 2018). The mel-spectrograms are then transformed to the decibel scale and a min-max normalization is applied over each one. The size of the resulting mel-spectrograms is $F \times T$, where $F = 64$ and $T = 251$ are, respectively, the frequency and time dimensions of the input to both, the supervised and self-supervised learning systems. Figure 1 shows four mel-spectrograms corresponding to each one of the classes under consideration.

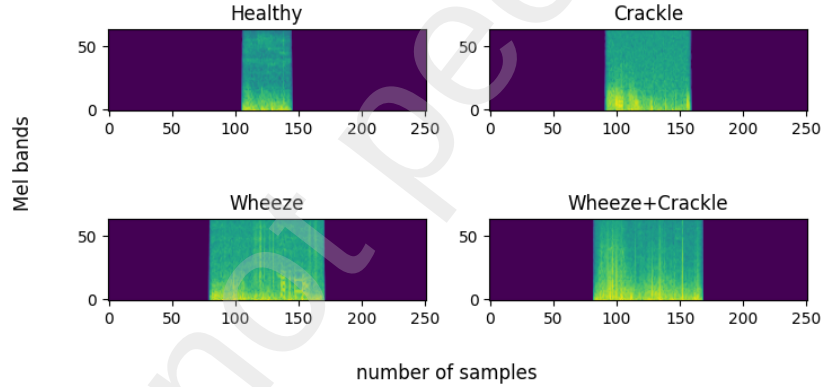


Figure 1: Mel-spectrograms with zero padding of four audio samples belonging to each one of the four types of lung sounds: normal, crackle, wheeze and wheeze+crackle.

3.2. *Class-imbalanced data*

As mentioned in Section 2, the ICBHI dataset, as typical in medical data, presents a skewed distribution of the classes, making it difficult to find a model that fits all categories rather than being biased towards the majority

class, that in this case, corresponds to normal lung sound. The literature lists several methods in related works to overcome this hamper that can be mostly grouped into class-balancing losses (Cui et al., 2019) and synthetic data-balancing methods (Chawla et al., 2002). Among these techniques, in this paper we have selected the two following ones due to their reported good performance: Adaptive Weighted Loss from the first group, and Weighted Random Sampler from the second one. Both are based on the computation of class weights by normalising the inverse of the relative frequencies of the categories in the training set, assigning higher weights to tail classes and a lower weights to head ones.

In the first technique, the class weights are incorporated into the training process through a weighted loss function that is the objective to be minimized. In this case, the contribution to the loss of a training instance is proportional to its weight that is the one corresponding to the class to which it belongs. The second strategy is a sampling method where the class weights are used for the building of balanced training batches, in such a way that the training instances are assigned a frequency of appearance in the batch given by their weights. This latter approach might suffer from a typical problem of sampling strategies: lost information from the majority classes due to samples not present in every epoch, and repeated information from multiple appearances of instances belonging to the minority classes. To solve the issue of redundant samples, data augmentation is performed to generate different versions of the same instance. Specifically, we have applied the SpecAugment technique (Park et al., 2019) on the mel-spectrograms (see Subsection 3.3).

3.3. Data Augmentation

In this work, we have used a variety of data augmentation techniques for several purposes. Firstly, as aforementioned, for solving data imbalance, we have employed SpecAugment in the training stage with Weighted Random Sampler of the supervised learning system and in the fine-tuning step of the self-supervised learning approach (see Subsections 3.4 and 3.5, respectively), i.e., when labeled data is involved. Secondly, to study the effect of the type of data augmentation on the generation of self-supervised representations (see Subsection 4.4), we have considered the following alternatives: SpecAugment, Noise Injection, Time Stretching, Vocal Tract Length Perturbation (VTLP) and a combination of several transformations (RawAugment). All the augmentations are applied directly on the raw audio signals but SpecAugment. In the following, we give some details about these methods:

- SpecAugment: This augmentation technique was proposed in (Park et al., 2019) and consists in applying one or more of the following modifications directly to the mel-spectrograms: (1) time and/or frequency warping; (2) frequency masking; and (3) time masking. In our case, the first one is not considered. The masking procedure is done with stripes along the frequency and/or time domain. The stripes, whose width is randomly generated from a uniform distribution from 0 to a predefined maximum value, are located at random starting points. For example, in the time domain, a stripe of width τ with starting point t_0 would apply a mask to the time interval $[t_0, \min(T, t_0 + \tau))$, where T is the total duration of the signal. This augmentation accepts a generation of multiple stripes for the different domains. For our study, both time

and frequency domains are affected by two stripes each. The maximum width values for time and frequency stripes are set to 50 and 20, respectively.

- Noise Injection: The augmentations within this category directly add some noise to the original audio signal. We have considered two versions:
 - White Gaussian Noise: In this case, a Gaussian noise of a certain amplitude is injected to the signal. The amplification factor depends on a minimum and a maximum amplitude range, that in this case is $(10^{-5}, 5 \cdot 10^{-5})$. It was deemed necessary to reduce both values since higher values might completely mask the original signal due to its low amplitude.
 - SNR Noise: This augmentation also adds Gaussian noise to the signal. However, contrary to the previous one, it does according to a Signal-to-Noise ratio (SNR) target, lowering the chances of a complete masking if the range of allowed SNRs is properly chosen. In this case, the minimum and maximum SNR values are set to 5 dB and 40 dB, respectively.
- Time Stretching: It changes the speed or duration of the signal without changing the pitch. Under the hood this augmentation is performed by means of a phase vocoding, so that the audio quality might be degraded. The process is carried out with a rate parameter r . If r is greater than 1, the signal is speeded up by a $100(r - 1)\%$, otherwise it is slowed down by a $100(1 - r)\%$. The defined range of parameter r is

(0.8, 1.2).

- Vocal Tract Length Perturbation: Vocal Tract Length Normalization (VTLN) was originally used in speech-related tasks to remove the speaker variability factor that deviates the specific speaker vocal tract length from the canonical mean. However, (Jaitly and Hinton, 2013) proposed to alter this method for adding speaker variations by warping the frequency axis according to a random warp factor α . In this case, the warp factor ranges between 0.9 and 1.1.
- RawAugment: Employed in (Soni et al., 2022), this augmentation encompasses several transformations to generate a more complex one. They are carried out in the following order:
 - Gain: It multiplies the audio signal by a random amplitude factor to increase or decrease its intensity. The amplitude factor range is between -12 dB and 12 dB .
 - Pitch Shift: It shifts the audio signal pitch up or down without tempo changes. This transformation uses phase vocoding so an audio quality degradation may occur. The audio is pitched by a factor that indicates the number of semitones. If it is negative, it shifts down, otherwise it shifts up. The range of this factor is between -4 to 4 semitones.
 - Temporal Shift: The signal is shifted forward or backward by a random fraction of the total sound length. In our case, this transformation does not do a rollover in case the shift reaches

one of the extremes of the duration. The fraction to be shifted is between -0.25 and 0.25.

- Time Stretching: Previously defined augmentation, but in this scenario the stretching factor is in the range between 0.9 and 1.1.

3.4. Supervised learning for respiratory sounds classification

The Supervised Learning system that serves as the baseline model for this study is inspired on the CNN6 architecture proposed in (Kong et al., 2019). As can be observed in Figure 2, it consists of a CNN deep network that acts as a feature extractor (block named “*CNN Feature Extractor (Encoder)*” in this Figure), followed by a set of stacked fully-connected neural networks, that performs the classification process itself (block named “*Supervised Learning*” of the upper branch).

In the convolutional part, the information is processed in a hierarchical way through 4 convolutional blocks, each of them consisting of a 2D-CNN layer with 5x5 kernels followed by a Batch Normalization layer, a non-linear activation function (in this case, a ReLU) and a 2x2 average pooling for dimensionality reduction. The number of filters used in each CNN layer is 64, 128, 256 and 512, respectively. To vectorise the output of this convolutional part, a sequential pooling is performed in both time and frequency dimensions until a vector with dimension matching the number of filters of the last convolutional layer (512, in this case) is obtained. Assuming a 1-sample batch for easier understanding, this global pooling operation is as follows:

Given $x \in \mathbb{R}^{K \times F \times T}$, where K , F and T are the number of filters in the last convolutional layer, frequency and time dimensions respectively, the first operation of the sequence is getting $x_T \in \mathbb{R}^{K \times F}$ calculating the average over

time. We later compute $x^{max} \in \mathbb{R}^K$ the maximum value of x_T over the frequency dimension and $x^{avg} \in \mathbb{R}^K$ the average value of x_T over frequency too. Lastly, the output of the global pooling is the sum $x^{max} + x^{avg}$.

As for the classification part, the original CNN6 model includes two fully-connected layers after the convolutional blocks. In our case, we do also add two dense layers but with different output dimensions compared to that presented in (Kong et al., 2019) given the difference in the corresponding tasks. In particular, the first layer is composed of K nodes (512, in this case), whereas the output dimension of the last one matches the number of sounds classes of interest, which is 4.

In this supervised approach, both blocks are jointly optimized by means of the minimization of a cost function that requires the availability of annotated training data. In this work, the selected loss function is the well-known multi-class Cross-Entropy (CE) which is defined in Eq. (1):

$$\mathcal{L}_{CE}(x, y) = - \sum_i^C y_i \log \frac{\exp(x^i)}{\sum_j^C \exp(x^j)} \quad (1)$$

where x is the logit vector, y_i is the true label taking a value of 0 or 1 depending if y class is i or not and C the total number of classes.

3.5. Self-supervised contrastive learning for respiratory sounds classification

One of the problems with supervised learning models employing deep learning networks is the high need for labeled data (and therefore, for computational resources) for smooth performance, as opposed to traditional machine learning models. In addition, they exhibit greater capabilities through non-linear construction relationships at the cost of a greater predisposition

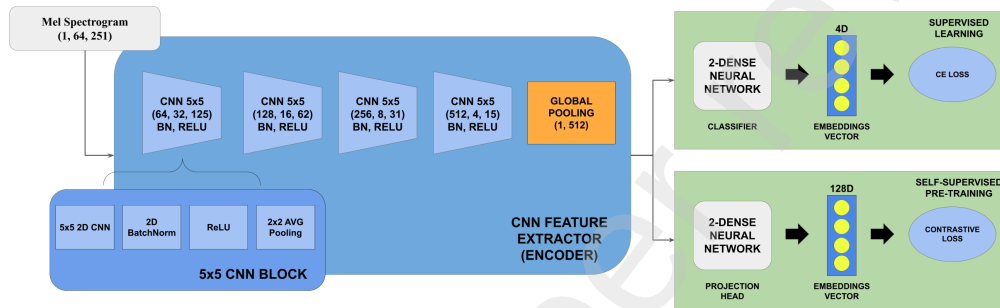


Figure 2: Block diagram of the supervised (upper branch) and the self-supervised pre-training (bottom branch) strategies. As can be observed, both approaches share the same convolutional architecture that serves as feature extractor. In the case of SL, both the convolutional and classification blocks are jointly trained by minimizing the multi-class cross-entropy loss function, being the output of the whole system a vector of probabilities assigned to each of the four classes under consideration. In the case of SSL, the block diagram corresponds to the pre-training stage where the whole architecture is optimized by minimizing a contrastive loss function, being the output an embedding vector of 128 dimensions that represents an audio recording and it can be further transferred as input to a subsequent downstream task.

to overfit the data rather than learning its general trends.

Another drawback present in supervised learning models is its tendency to bias towards majority classes when using loss functions such as cross-entropy, where, by design, the value of the gradient is higher for majority classes favouring the learning of these classes and, therefore, reducing the model's "attention" to detect patterns or characteristics of classes that are less present in the data.

A new paradigm in DL, called self-supervised learning, has recently emerged, and has been gaining some popularity in the scientific community due to its features that compensate for the problems present in supervised models, and their promising results obtained. The main idea behind this framework lies in addressing a *pretext task* to train a model able to generate valuable features or representations without the utilization of labeled data. Then, these representations can be later employed in other problems of interest called *downstream tasks*. The process of pre-training a model using self-supervision is often called Self-Supervised Pre-training (SSP) in the literature. Pretext tasks are not necessarily directly related to the target downstream tasks. Some examples are guessing properties of original input data (i.e. predicting image orientation in computer vision) or predicting masked information (i.e. filling masked/removed segments of an audio or speech). Once the model (usually referred as pre-trained model, encoder or backbone) is learnt on the pretext task without annotated data, it is adapted to the downstream task by aggregating a sub-architecture suitable for it (e.g., several dense layers acting as a classifier) that is trained accordingly in a supervised fashion. The training of the downstream task can be done by partly or entirely freezing the

architecture of the pretext task or fine-tuning the whole model. Although it is commonly established that the pre-training stage typically necessitates a significant amount of unlabeled training data, one of the main advantages of this approach is that the adaptation process to the downstream task requires only a limited amount of labeled data.

In our study we have focused on the branch based on contrastive learning, inspired in the dimensionality reduction technique proposed in (Hadsell et al., 2006). This strategy seeks to learn of a transformation that maps the input data to a representation space, in such a way that preserves the relationships present in the input space. This way, similar objects or *positive pairs* in the input space would have representations with a lower distance than those from dissimilar objects or *negative pairs*. In contrast to the CE loss of the supervised approach, this approach uses the InfoNCE loss (van den Oord et al., 2018), that is based on the Noise Contrastive Estimation (NCE) loss and adds a temperature parameter for controlling the penalty on the effect of negative samples. The InfoNCE loss is defined as,

$$\mathcal{L}_{\text{InfoNCE}}(x) = -\log \left(\frac{\exp(\text{sim}(x, x_+)/\tau)}{\exp(\text{sim}(x, x_+)/\tau) + \sum_{x_- \in \mathcal{X}^-} \exp(\text{sim}(x, x_-)/\tau)} \right) \quad (2)$$

where x is the sample to compute the loss, x_+ is the positive pair of x , x_- are the negative pairs of x and sim is the similarity distance in the output space. The commonly used similarity distance is the cosine distance defined according to Eq. (3),

$$\text{sim}(x, y) = \cos(x, y) = \frac{x^T y}{\|x\| \cdot \|y\|}. \quad (3)$$

In this paper, we have adopted for our SSL system the SimCLR technique (Chen et al., 2020), a simple contrastive learning framework which has gained quite popularity from its simplicity and good performance. SimCLR builds the feature representation space by defining the similarities according to different views of samples generated via data augmentation. More specifically, the aim of the pretext task considered for training the encoder consists of pulling together views generated from the same sample and pulling it apart from the views from different samples. The positive samples are the different data augmented views of a random sample x while the negative samples are the augmented versions of all samples but x . Note that the data augmentation procedure is employed for generating pseudo-labels (positive and negative samples) and therefore, labeled data is not required. SimCLR defines a neural network *encoder* f , that extracts *representations* from the original input space producing the so-called *representation space*. Instead of applying the InfoNCE loss in the representation space, as usual in other CL variants, the SimCLR model adds a neural network *projection head* g that returns the *embeddings* of the representations generating the *embedding space*. In (Chen et al., 2020) it is shown that applying the loss in the embedding space instead of directly in the representation one improves the quality of the learned representations when evaluated by means of a simple linear classifier. The SimCLR process is depicted in Figure 3.

In our SSL-based system, we have employed SimCLR to pre-train a CNN encoder architecture that would later generate the features or representations to be forwarded to our later classifier in the automatic respiratory sound classification downstream task. The CNN architecture is the same as the one

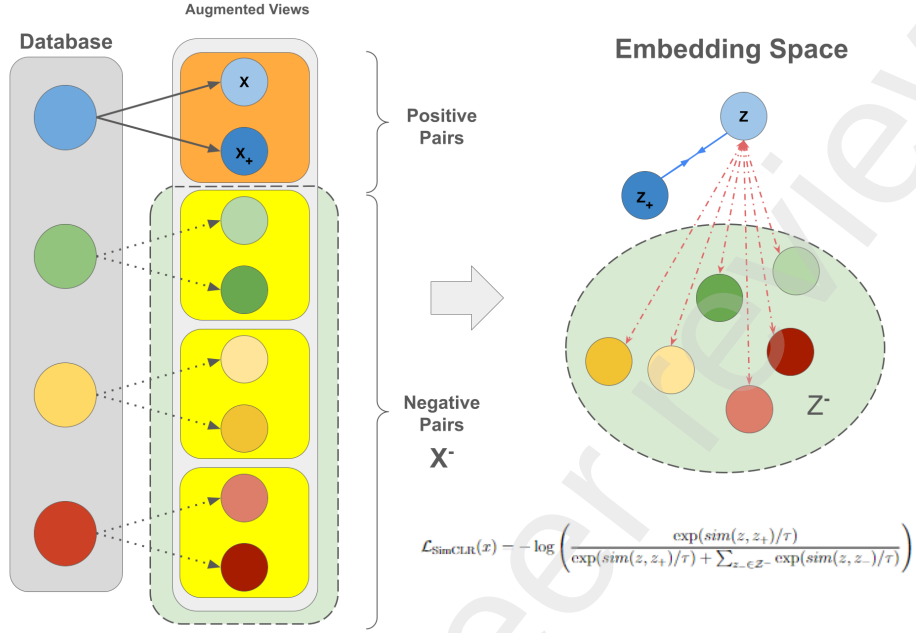


Figure 3: Diagram of the generation of augmented views: both positive and negative pairs. For each sample in the original database two views are generated using the data augmentation considered. The augmented views that come from the same original sample are considered positive pairs, and those that come from a different sample are negative pairs. In SimCLR, the InfoNCE loss is applied in the embedded space. Given an encoder f and a projector g , an augmented view x is represented by the embedding vector $z = g(f(x))$. In the diagram we view the InfoNCE contrastive loss respective to the embedding of the augmented view x . It can be appreciated that x_+ is an augmented view of the sample from where we get x so they are a positive pair, while the others views belong to the set of negative pairs of x (X^-). In the embedding space, z and z_+ are the positive pairs we aim to pull together, while aiming to pull apart from z all the embeddings of its negative pairs (Z^-).

considered in the convolutional part of the fully supervised system as shown in Figure 2, ensuring a fair comparison between both. From this Figure, it

can be observed that during the SSP process, for each input mel-spectrogram the encoder returns a 512 dimensional vector that is later processed by a 2-dense neural network projector head. In our case, the projector outputs a 128 dimensional vector embedding vector. The embeddings of the positive and negative pairs built following the procedure shown in Figure 3 are introduced in the contrastive loss defined in Equation (2) with a temperature of $\tau = 0.07$. The encoder is trained by minimizing this loss function in the usual iterative fashion.

Once the encoder is generated through the pretext task using SimCLR, it can be used in the downstream task, following the procedure depicted in Figure 4. As shown in Figure 4.A, after the SSP process is finished, the projection head is removed, keeping the pre-trained encoder that is transferred to the training step of the downstream task according to 4.B. In this stage, a 2-dense neural network working as a classifier is attached to the output of the pre-trained encoder, whose weights are used for the initialization of the whole model. This model is subsequently trained with the available labeled data in a supervised way using the cross-entropy loss. This training process can involve either partially or entirely freezing the encoder’s architecture or fine-tuning the entire model. Note that in this scheme, the classifier has the same architecture as its counterpart of the fully supervised system.

4. Experimental results

4.1. Dataset splitting strategy

As mentioned in Section 2, following the experimental protocol established in the ICBHI challenge, the original dataset was manually segmented

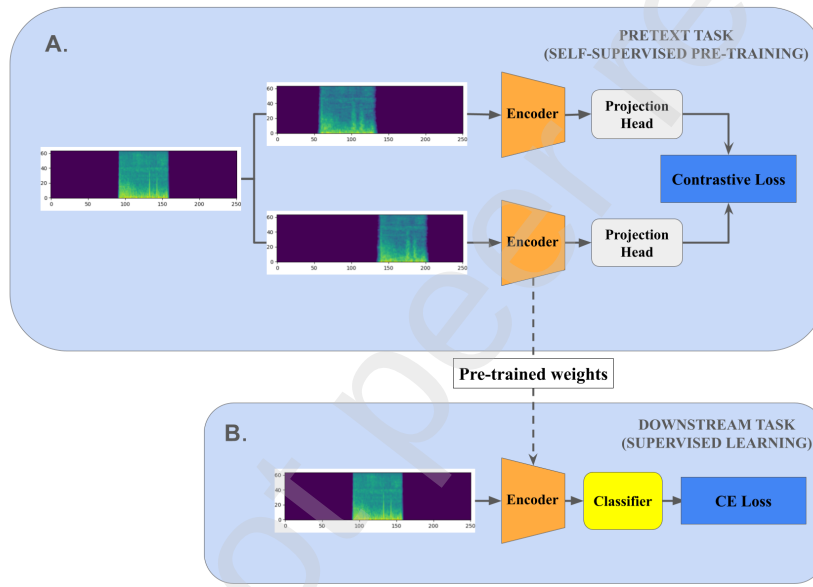


Figure 4: Block diagram of the self-supervised-based system. **A.** Self-supervised pre-training process of the pretext task using SimCLR. **B.** Supervised learning of the downstream task that can involve either partially or entirely freezing the pre-train encoder or fine-tuning the entire model.

into respiratory cycles. For that reason, in the final dataset, there are several audio files belonging to the same patient. This fact could produce data leakage from the training to the evaluation sets if the dataset is not properly splitted. To avoid a training bias towards detecting characteristics and patterns inherent to patient instead of the semantics of the respiratory sounds of interest, a partition of the dataset by patient with all its cycles has been made. This prevents the existence of recordings from the same subject in different training/test sets.

These single-patient subsets are arranged into 5 disjoint groups suitable for performing a 5-fold cross-validation evaluation of the models. In each fold, three groups are utilized for training, another different group is used for validation, whereas the remaining one is kept for testing.

4.2. Evaluation criteria and bootstrapping

The measures used for the evaluation of the different systems are similar to those proposed in the 2017 ICBHI challenge. In particular, the metrics employed in the downstream task (classification of the 4 different lung sounds) are:

$$\begin{aligned} Se &= \frac{P_d}{T_d}; & Sp &= \frac{P_n}{T_n} \\ Sc &= \frac{Sp + Se}{2}; & Acc &= \frac{P_d + P_n}{T_d + T_n} \end{aligned} \quad (4)$$

where Sp and Se are, respectively, the specificity and sensitivity considering the categories related to an abnormal lung sound as the class positive and as negative class, the healthy or absence of respiratory anomaly, Sc is the

score metric defined at the ICBHI challenge, and Acc is the accuracy of the model. Taking these metrics into consideration, P_i is the number of samples predicted to belong to class i and T_i is the number of ground truth samples in class i , where sub-index d refers to abnormal sounds (wheezes, crackles and both) and n refers to normal or healthy respiratory sounds samples.

In order to provide a more holistic view of the model performance, the multi-class Area Under the Curve (AUC) has been used as this metric integrates the true positive and false positive rates in a single value, and it is less sensitive to imbalanced datasets compared to accuracy. This measure can be computed by considering the macro-average of the true and false positive rates when the output of the classifier is binarized to distinguish, respectively, a certain class from the rest (OneVsRest, OVR) or a pairwise combination of classes (OneVsOne, OVO). In order to consider both points of view while maximizing them at the same time, the minimum between the macro-mean OneVsRest and the macro-mean OneVsOne has been taken as the metric, as shown in Equation (5).

$$\delta AUC = \min\{AUC_{macro}(OVR), AUC_{macro}(OVO)\}. \quad (5)$$

Results achieved by deep learning methods present a certain degree of variability due to a set of random elements that participate in their training mechanism: from the random initialisation of the weights, to the random sampling of the training batches to obtain the estimation of the gradient. For that, it is deemed necessary to compute confidence intervals for the performance of the models in the Subsections where comparisons are made (see Subsections 4.5 and 4.6).

Considering the computational burden of training the same model and evaluating it several times to obtain a distribution of its performance and respective metrics, it has been preferred to bootstrap the evaluation of the test set given the trained model. For this purpose, a bootstrapping resampling with replacement has been applied to the test set until samples matching the test set's size have been obtained, and the metrics of interest have been calculated for each sample. This process has been performed 400 times to obtain reliable distributions of the evaluation measures.

As a consequence of adopting a 5-fold cross validation strategy in the experimental protocol, the bootstrapping method has been carried out taking the different folds into account. In particular, in each iteration of the bootstrapping process, we did a random resampling with replacement for every fold independently, later computed the metrics for the resamples from every fold and finally performed the macro-average of the metrics of all folds for the 400 different resamples. Results reported in this Section correspond to the average value from all different bootstrapped samples macro-average across the 5-fold subexperiments.

4.3. Results of the supervised learning system

First, we proceed to analyse the results obtained over the downstream task (respiratory sound classification) when generating a model with supervised learning from scratch. These results show us the capacity of the baseline system and allow the comparison with models trained with the self-supervised learning strategy.

As mentioned before, the dataset is imbalanced with a clear bias towards the normal class, so in order to alleviate this issue, the two solutions pre-

Balance Strategy	Metrics				
	Specificity	Sensitivity	Score	Accuracy	δ AUC
None	0.8139	0.4104	0.6122	0.6226	0.7626
Weighted Loss	0.7942	0.4485	0.6214	0.6290	0.7811
Weighted Sampler	0.8334	0.4099	0.6216	0.6333	0.7889

Table 1: Results of the supervised learning system with different strategies for addressing unbalanced data.

sented in Section 3.2, Adaptive Weighted Loss (Weighted Loss) and Weighted Random Sampler (Weighted Sampler), have been studied. The Weighted Sampler strategy utilizes a SpecAugment data augmentation for the generation of balanced batches. In both approaches, models have been trained using a batch size of 32 and 100 epochs. The loss function is the multi-class cross-entropy defined in Equation (1) that has been optimised by employing the Adam algorithm with betas 0.9 and 0.99, learning rate 10^{-3} and weight decay 10^{-4} . For controlling the learning rate, a cosine annealing scheduler has been added with the same maximum number of iterations as the epochs used in training.

As can be seen in Table 1, on the one hand, both class-balanced approaches obtain better score, accuracy and δ AUC than when no balance strategy is used. On the other hand, Weighted Sampler outperforms Weighted Loss in terms of specificity in exchange of a lower sensitivity. Nevertheless, it obtains slightly better outcomes looking at score, accuracy and δ AUC. For this reason, from now on, all models trained in a supervised way will be obtained using this strategy.

As for the comparison with other state-of-the-art works, we get compa-

rable results to the compilation of studies in (Petmezas et al., 2022), when a patient-independent K-Fold cross validation configuration similar to ours is used. Note that a direct comparison is not feasible due to the lack of public distribution of the recordings per fold.

4.4. Results of the self-supervised learning system: Effect of data augmentations on self-supervised representations

In the self-supervised framework, several studies have verified experimentally that there is a relationship between model quality and the data augmentation technique used to generate the views of the input objects. Moreover, in (Jing et al., 2021) it is theoretically presented how the change in the variability of the data generated by applying such techniques with respect to the original dataset influences the dimensionality of the representation space, which involves a difference in the information available to the classifier.

These works typically deal with image-related problems, in such a way that limited studies have been done on audio processing tasks, highlighting (Al-Tahan and Mohsenzadeh, 2020) where the influence of the type of augmentation in the quality of the audio representations have been analysed for speech commands recognition and music and environmental sound classification. Concerning lung sounds, (Soni et al., 2022) realizes a similar study for the task of discriminating healthy patients from non-healthy ones on the ICBHI dataset. However, to our knowledge, this phenomenon has not been previously researched for the multi-class classification problem addressed in this paper. To fill this gap, and focusing on our task, in this set of experiments, we have generated several models by means of self-supervised contrastive learning using different data augmentation techniques.

In particular, we have chosen the alternatives presented in Section 3.3 as are commonly used in auditory tasks with deep learning.

In the pre-training stage of the self-supervised learning, all configurations use a batch size of 128 and a LARS optimiser (You et al., 2017) with an initial learning rate of 0.2 and a weight decay of 10^{-6} for 100 epochs. For the learning rate, a linear increasing warmup in the first 10 epochs followed by a cosine annealing scheduler without restarts is applied. As stopping criteria, the pre-trained model (encoder) that produces the highest δAUC over the validation subset is selected.

These self-supervised models have been evaluated in two different scenarios. The aim of the first one (referred as *KNN*) is to determine the quality of the representation space generated by the pre-trained encoder. For doing this, a KNN model with 15 neighbors has been trained over these representations in a supervised fashion for the task of multiclass lung sound classification. The idea behind this scenario is that if the self-supervised model has been able to capture semantic features of the classes, it is expected that the distribution of the data representations in this space will be concentrated around the classes and good results will be attained with such a simple classification model and with the weights of the encoder frozen.

The objective of the second scenario (referred as *Fine-Tuning*) is to evaluate the performance of the self-supervised representations in the considered downstream task, i.e., classification of lung sounds. In this case, as indicated in Figure 4.B, a deep-learning classifier following the architecture of the classification part of the supervised system (i.e. two fully-connected layers with, respectively, 512 and 4 nodes) is attached to the self-supervised encoder with

Data Augmentation	Task	Metrics				
		Specificity	Sensitivity	Score	Accuracy	δ AUC
SpecAugment	KNN	0.7707	0.3023	0.5365	0.5475	0.6567
	Fine-Tuning	0.8078	0.4468	0.6273	0.6352	0.7910
Gaussian Noise	KNN	0.7528	0.3130	0.5329	0.5427	0.6647
	Fine-Tuning	0.8530	0.4013	0.6271	0.6426	0.7938
SNR Noise	KNN	0.7455	0.3244	0.5350	0.5449	0.6798
	Fine-Tuning	0.8010	0.4285	0.6148	0.6252	0.7856
TimeStretch	KNN	0.7367	0.3589	0.5478	0.5562	0.6952
	Fine-Tuning	0.8215	0.4401	0.6308	0.6401	0.7902
VTLP	KNN	0.7428	0.3156	0.5292	0.5398	0.6738
	Fine-Tuning	0.7942	0.4307	0.6125	0.6222	0.7892
RawAugment	KNN	0.7492	0.3303	0.5398	0.5493	0.6881
	Fine-Tuning	0.8054	0.4396	0.6225	0.6347	0.7937

Table 2: Comparative results between different data augmentations used for self-supervised pre-training under the two considered scenarios: *KNN* and *Fine-Tuning*.

the best weight configuration determined in the pre-training stage. This global system is subsequently fine-tuned in a supervised way using the same training dataset, hyperparameters and Weighed Random Sampler strategy as in the fully supervised model.

In (Al-Tahan and Mohsenzadeh, 2020), it was suggested that combining sequentially data augmentation brings better performance to the self-supervised models for the tasks of speech commands, music and environmental sound classification. Looking at our results in Table 2, this observation could not be confirmed, as it might be appreciated that RawAugment does not get the best performance, although its metrics in both *KNN* and *Fine-Tuning* scenarios are close to the best ones. In the *KNN* case, TimeStretch

outperforms the remaining methods in terms of score, accuracy and δAUC . In the *Fine-Tuning* scenario, it is remarkable the good performance of a simple augmentation like white Gaussian Noise as it achieves the best results for accuracy and δAUC followed very closely by TimeStretch, that obtains the best score value. As for the comparison between both scenarios, it can be concluded that fine-tuning the model in a supervised manner notoriously increases the performance in all metrics as expected.

Lastly, taking into account that TimeStretch is the best method in terms of sensitivity and the results achieved in the remaining metrics, this data augmentation is used for the training of the self-supervised models in Subsections 4.5 and 4.6.

4.5. Results of the self-supervised learning system: Effect of the training data size

One of the reasons for the growing interest in self-supervised learning models derives from the premise that they are more suitable for databases with a smaller amount of labeled data by allowing the model to be pre-trained without requiring such labels and subsequently refined with the available annotated data. In order to evaluate this hypothesis for our case, we have compared the performance of the fully supervised learning model with the self-supervised contrastive learning system with fine-tuning by carrying out several subexperiments considering different amounts of labeled data from our database. On the one hand, since the supervised learning model depends on the existence of labels, it will only be trained with the subset of labeled data corresponding to each subexperiment. On the other hand, since the pre-training stage of self-supervised system does not need labels, we will obtain

the pre-trained model with the whole training data and then perform a fine-tuning using only the subset of annotated data of each subexperiment as in the case of the supervised system.

In each of the subexperiments of this Subsection, a proportion of training labeled instances is discarded as missing values until it is reached the desired proportion of data to be lost. To make it more in line with reality, two elimination strategies have been envisioned: by patient and by recording. In the first case, a proportion of patients is selected for which no labels are collected for any of their recordings (see Figure 5), whereas in the second case, a random subset from all recordings of patients are chosen to be discarded (see Figure 6). These two approaches align with the real scenarios where a doctor does not diagnose a patient, then discarding all the labeled information related to that patient (first strategy), and the second scenario is when given a patient with several audio recordings, only a few of them might have been annotated. The procedure finally selected and set as default is the first one.

Figure 7 shows that both systems exhibit similar results the closer we consider the 100 % labeled data proportion, suggesting that both models can perform comparably when given a similar amount of labeled information to exploit. Nevertheless, SSL slightly increases the results of SL producing a relative improvement in score, accuracy and δAUC of respectively 1.2%, 0.8%, 0.4% with respect to SL. It is particularly interesting that, when all the available labeled data is employed, SSL statistically outperforms the conventional approach in terms of sensitivity achieving a relative improvement of 11.3%. In this case, SSL presents a relative specificity reduction of 3.7%

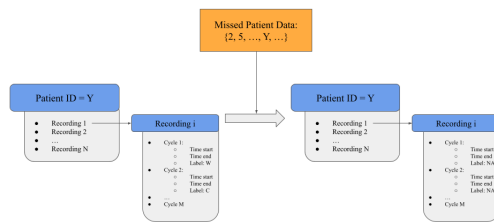


Figure 5: Schema of the elimination strategy by patient. NA stands for “Not Available” when the corresponding label is discarded.

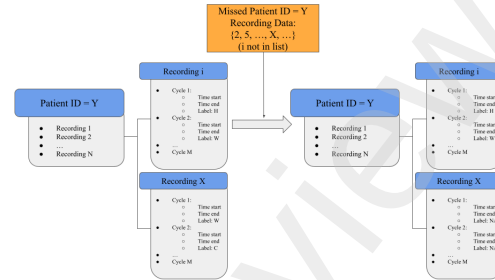


Figure 6: Schema of the elimination strategy by recording. NA stands for “Not Available” when the corresponding label is discarded.

with respect to SL, although this difference in performance is not significant.

Nevertheless, in the range from 30% to 55% of labeled data, the self-supervised system shows better classification power, as it is able to leverage unsupervisedly the unlabeled data during the pre-training stage, and successfully transfer this information to its fine-tuning stage. Moreover, looking at both sensitivity and specificity while considering the other metrics, SSL displays a lower biased effect to the majority class (a statistically better sensitivity outcome) while getting statistically higher values in score, accuracy and δAUC . In particular, when the labeled information is reduced by 50 %, SSL achieves a relative improvement in score, accuracy, δAUC and sensitivity of respectively 5.8%, 3.3%, 8.1% and 74.2% with respect to SL, in contrast to a 10.9% of relative specificity reduction. The confusion matrices for both systems depicted in Figure 8 help to further analyse this phenomenon. As can be observed, the accuracies of the minority classes (wheezes, crackles and both) achieved by SSL surpass those obtained by SL when the 50% of

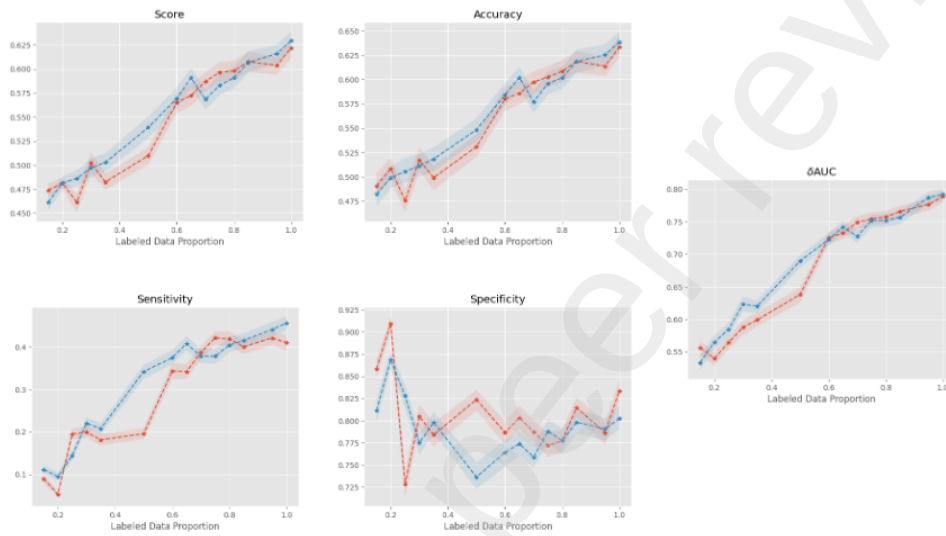


Figure 7: Results of the baseline supervised system (**red**) vs. the fine-tuned self-supervised system (**blue**) as a function of the proportion of labeled training used. All curves represent mean (dots) and the confidence intervals of each subexperiment computed with the bootstrapping methodology explained in Section 4.2.

the labeled training data is used, although the classification rate of the head class (normal sound) is lower. When adding more labeled data to the training process, all accuracies improve, including that of the majority class. In contrast, with SL, there is a clear bias towards the majority class when using 50% of the labeled data, although this bias is mitigated when the entire labeled training data is employed.

Therefore, in these circumstances, SSL is able to get profit of the unlabeled data for building a representation space that can be refined during the fine-tuning stage with a limited amount of annotated data. However, the performance of the supervised system degrades due to the labeled data scarcity, especially in terms of sensitivity, score and δAUC .

Finally, when the available label data is too much scarce (less than 30% of annotated exemplars), both models downgrade considerably and there is no clear conclusion around which model performs better. Given our skewed distributed small database, when the proportion of labels is very scarce there is a quite limited amount of samples from the minority classes, which would induce the need of considering adding a few-shot learning approach in that scenario.

4.6. Results of the self-supervised learning system: Analysis of the representation space

As previously stated, supervised cross-entropy learning often generates poor separability margins between classes when exists a substantial imbalancing factor, i.e. the ratio of samples in the head classes with respect to the other classes is high. In this case, the representations of the classes are distributed in a biased manner, thus generating a space where dominant classes



Figure 8: Confusion matrices of the supervised learning (left) and self-supervised learning (right) systems for the cases where the 50% (top) and 100% (bottom) of the labeled training data is used. The confusion matrices are normalized by the sum of each row.

occupy larger areas. (Kang et al., 2021) studies this phenomenon comparing the representation spaces of SL and SSL in a computer vision task and defines a balancedness metric, where given a set of representations of a space and a classifier, it measures the overall differences between the accuracies of the different classes. A balanced representation space is expected to distribute the classes equally with similar separability margins, encouraging the model to get similar accuracies for all classes. This balancedness metric considers a squared distance and all pair-wise combinations of classes accuracies. The closer the accuracies between classes are, the higher the value of the metric is, being 1 the maximum possible value. We have modified the original definition of this metric by using an absolute distance (instead of the euclidean one) and pair-wise combinations without repetition to avoid repetition of contribution of pairs to the metric as shown in Equation (6). In particular, given a C number of classes with a_i corresponding accuracy, the modified metric is as follows:

$$\mathcal{B} = \frac{1}{\hat{C}} \sum_{j>i}^C \exp\left(-\frac{|a_i - a_j|}{\sigma}\right), \quad \text{where} \quad \hat{C} = \frac{C(C-1)}{2} \quad (6)$$

As can be observed, we have changed the denominator of the original definition of the metric in (Kang et al., 2021) from C^2 to the number $\hat{C}$ of pair-wise unique combinations without repetition. Excluding the pair-wise repeated combinations we relieve the minimum bias possible value of this metric of $1/C$. There is also an implicit bias towards the minimum value of this metric when we increase the number of classes, that can be reduced lowering the value of σ in Equation (6). However, since the comparisons in our study have the same number of classes, the specific value of σ does not

affect the relative comparison between the balancedness of the different cases under study.

In contrast to the work of (Kang et al., 2021), we do not modify the imbalancing factor of our database, since it would alter the *a priori* frequencies of the medical classes of the study. However, keeping in mind that the main focus of our work revolves around deep learning models used in biomedical applications, it is interesting to study the evolution of the balancedness of both, the SL and SSL systems, with respect to the number of labeled data available, which is one of the critical aspects of biomedical applications.

Figure 9 shows that both models experiment statistically similar performance when the amount of labeled data is in the range from 60% to 80%. However, SSL registers a sudden increase of the balancedness value in the range from 30% to 55%, where in Subsection 4.5 it was observed that the evaluation metrics (except for Specificity) showed a better classification power coming from this model (see Figure 7). Taking into account previous and current results, it can be concluded that SSL prioritizes achieving a balanced representation state before focusing on extracting more discriminative patterns for classification. This is evidenced by SSL reaching the final balanced state value of SL with significantly less labeled data, around 50%. From that point onward, as more labeled data is incorporated, the algorithm concentrates on enhancing its predictive capabilities while maintaining the balanced state of the space. This idea is further reinforced by the observation that SSL sacrifices some specificity in the head class performance to achieve a much better balanced model. This results in a slight increase in overall model performance and a significant improvement in balancedness and sen-

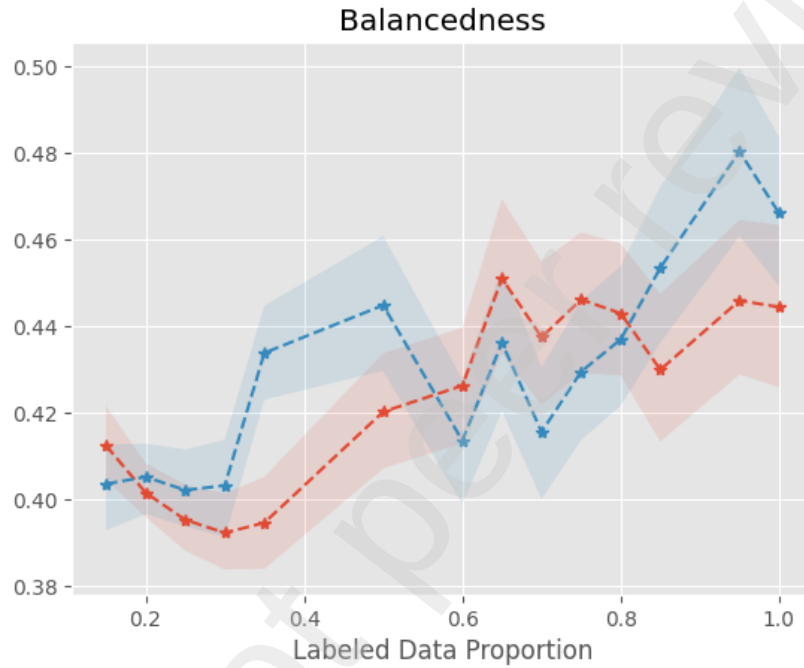


Figure 9: Balancedness of the baseline supervised system (**red**) vs. the fine-tuned self-supervised system (**blue**) as a function of the proportion of labeled training used. The balancedness metric is calculated with $\sigma = 0.5$. All curves represent mean (dots) and the confidence intervals of each subexperiment computed with the bootstrapping methodology explained in Section 4.2. It can be appreciated that the balancedness value of supervised model with whole labeled data proportion is reached by self-supervised with much less labeled data, around 50%.

sitivity. This improvement can be attributed to the model’s enhanced ability to address bias towards majority classes, thereby boosting the performance of tail classes. This conclusion correlates with the observations made by (Kang et al., 2021) in the image processing field, which states that SSL creates a more balanced space with less discriminant power, whereas SL learns a biased space but with more discriminative ability. Adding more label information when the space reached its maximum balanced state only leads to an improvement in the predictive and discriminative capabilities of the model.

4.7. Results of the VGGish pre-train model

In this work, we have obtained a pre-trained model leveraging the information contained in unlabeled data through SSL by using a database that is considerably small although related to our task of interest as it contains respiratory sounds. An alternative to ours relies on using open-access pre-trained models generated in a supervised way from large general-purpose external databases, but that either lack recordings of the acoustic classes of interest or contain only a minimal number of such recordings. In this context, our goal in this Subsection is to verify whether there is an advantage in using this kind of pre-trained models instead of SSL.

Specifically, we have chosen the well-known VGGish pre-trained model (Hershey et al., 2017) (Taylor et al., 2019) as it has been previously considered for our task in (Demir et al., 2020b). This model is based on the VGG convolutional architecture (Simonyan and Zisserman, 2015) adapting the system from image to sound, and it has been trained on a large Youtube dataset. It includes a preprocessing stage in which VGGish generates subsamples of the original audio sample by applying 1 second long analysis windows. These

subsamples are then transformed to mel-spectrograms and processed by the VGG network that returns a matrix of n 128 dimensional embeddings, where n is the number of subsamples generated with the preprocessing.

Since the VGGish preprocessing generates several subsamples for each original sound, we have established some rules for training and evaluation:

- During the training of the classifier, the subsamples are assumed to belong to the class of the complete original sample.
- The padding used is the circular one, as a zero padding would undermine the performance of the model due to the first rule, as there would be subsamples of zeroes assigned to different classes.
- In the evaluation of the model, the final probability of each sample is calculated across the mean of the probabilities of the individual subsamples.

As shown in Table 3, we have evaluated the VGGish-based system in two scenarios as in Subsection 4.4. In the first scenario (*KNN*), the embeddings from the VGGish model are the input to a KNN classifier. In the second case (*Fine-Tuning*), the VGGish model is attached to a classifier, that follows the same architecture that those used previously in the SL and SSL systems (see Figure 2), being the only difference that the hidden layer nodes changes to 128 to match the dimensionality of the VGGish output. In both cases, we have used VGGish as a feature extractor, in such a way that its weights are frozen during the training process. The hyperparameters employed in both scenarios have been the same like those used in the fine-tuning of the SSL

Training	Balance Strat.	Metrics				
		Specificity	Sensitivity	Score	Accuracy	δ AUC
KNN	-	0.8433	0.3285	0.5859	0.5983	0.7246
Fine-Tuning	None	0.8948	0.2610	0.5779	0.5939	0.7315
	Weighted Loss	0.8781	0.2941	0.5861	0.6018	0.7328
	Weighted Sampler	0.3797	0.5181	0.4489	0.4452	0.6970

Table 3: Comparative VGGish results under the two considered scenarios: *KNN* and *Fine-Tuning*.

model, but changing the learning rate from 1^{-3} to 1^{-4} in the second scenario since it provided better results. Also, we have considered different balanced strategies: None, Weighted Loss and Weighted Sampler as defined in 3.2).

Looking at Table 3, it can be appreciated that the *Fine-Tuning* approach seems to degrade the performance of *KNN* contrary to what happened in Table 2. Moreover, Weighted Sampler gets a much worse performance compared to the other strategies. This might be caused by a lost of relevant information from using the masking applied by the SpecAugment augmentation in a small subsample. However, it is worth mentioning this technique reaches the best sensitivity across all configurations from all the study, but with a much worse specificity. Comparing these results with those in Table 2, it can be observed that *KNN* with VGGish embeddings is considerably better than *KNN* with SSL indicating that the VGGish embedding representations are good. However, in the *Fine-Tuning* scenario, SSL significantly improves VGGish performance, suggesting that the information contained in the VGGish embeddings is not effectively transferred to the downstream task, likely due to the high discrepancy between the acoustic data used for

training the VGGish model and the data of the downstream fine-tuning process. Thus, it seems apparent that the quality of the training data used for the generation of the pre-trained model, in terms of its appropriateness for the downstream task, is more critical than the quantity of the data.

Finally, we have performed another set of experiments where the VGGish weights were unfrozen for allowing a fine-tuning of the whole architecture like we did with the pre-trained SSL. However, results did not surpass those presented in Table 3, possibly due to the difficulties of properly training a large convolutional model with a small database. This outcome is consistent with the findings in (Demir et al., 2020b).

5. Conclusion

In this paper, we have explored the use of self-supervised contrastive learning, specifically SimCLR, for the task of respiratory sound classification, and made a fair comparison with the conventional supervised learning approach. Throughout the results, we have verified the potential of this technique in addressing two challenges present in DL applications in biomedicine: limited labeled data and significant class imbalance. In fact, the SSL strategy outperforms SL in scarcity annotated data conditions and improves the balancedness of the representation space, thus enhancing the sensitivity of the classification system.

Additionally, we have investigated several critical factors affecting the performance of our SSL system, such as the impact of various data augmentation on the quality of the pre-trained SSL encoder and the variation in system performance with respect to the amount of labeled data available. Fi-

nally, the SSL model has been evaluated against an open-access, pre-trained audio DL model, showing that, for our target task, the proposed approach outperforms existing models trained on extensive external datasets.

However, we have also identified some possible limitations within this recent paradigm. Despite the promising results of SSL in creating a balanced classifier, it did not lead to a substantial performance improvement compared to the fully supervised approach when the amount of available labeled information is equivalent. This observation aligns with concerns from some researchers regarding potential loss in discriminative power of the self-supervised strategy. Moreover, when the amount of annotated data is very scarce, both models performed very poorly. These two aspects open the line for future research: considering few-shot learning techniques in the field of SSL biomedical audio applications, and adding extra information in the creation of the representation space to increase its discriminative capabilities.

Acknowledgement

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